LOST & FOUND KNOWLEDGE BOARD (Navibyte)

MVP APP DEFINITION

SECI-Based Knowledge Management and Lost Item Recovery Platform

Prepared for the 7-Week MVP Sprint Delivery Plan

Core Roles: Admin | Staff | Student

---

1. App Overview

The Navibyte Lost and Found System (LFS) is a web application designed to transform subjective human observations into standardised digital assets. By integrating the SECI Knowledge Management model, the platform addresses the "Vocabulary Gap," "Inventory Drift," and "Identity Risk" inherent in traditional lost and found operations.

The MVP relies on a process-driven approach to data schema, fixed taxonomy, and a secure retrieval logic workflow to ensure lost items are accurately logged and safely returned to their rightful owners.

2. Target Users

User Type	Primary Goal	MVP Access
Finder (Public User)	Accurately report found items without relying on subjective text descriptions.	Item submission (Categorical Icon Grid, Fixed Colour Palette, Zone Dropdown), Safe Drop coordination.
Claimant (Public User/Staff)	Search for lost items and securely verify ownership without exposing personal information.	Item submission (Categorical Icon Grid, Fixed Colour Palette, Zone Dropdown), Safe Drop coordination.
Staff	Manage physical inventory, oversee handovers, and prevent Inventory Drift.	Dashboard access, Bin Mapping management, audit logs, and view Hidden Notes for verification.
Admin	Maintain platform governance, taxonomy structures, and security oversight.	All Staff access plus Categorical Logic management, lifecycle management triggers, and user access control.

3. Problem Statement

Traditional lost and found systems fail due to three critical breakdowns:

- **Vocabulary Gap (Tacit-to-Tacit Mismatch):** Finders provide subjective descriptions (e.g., "dark bag" vs. "navy backpack"), making it impossible for owners to accurately search for their items.
- **Inventory Drift:** The digital record of an item becomes desynchronized with its physical location, causing organisational chaos (e.g., an item is marked "Found" digitally but has already been returned physically).
- **Identity Risk:** Direct contact between finders and claimants exposes Personally Identifiable Information (PII) and increases the risk of fraudulent claims or theft.

4. Core Solution

The system utilises a structured Knowledge Management architecture to eliminate "human noise." It replaces free-text descriptions with a Fixed Choice model (Categorical Logic and Attribute Standardisation) and secures the handover process through Data Tiering (Public, Private, Administrative) alongside an encrypted Masked Chat module for Blind Verification.

4.1 How SECI Will Be Used

The SECI model structures the conversion of subjective observations into a reliable chain of custody.

SECI Phase	Meaning	How the System Uses It
Socialization	Tacit-to-Tacit knowledge sharing.	**Connection Setup:** Staff utilize the Dashboard as a shared space ("Ba") to observe trends. Users share intent via Public or Anonymous privacy states.
Externilization	Tacit knowledge becomes explicit	**Discovery & Initiation:** Subjective observations are codified via the Categorical Icon Grid and Fixed Colour Palette to eliminate the Vocabulary Gap.
Combination	Explicit knowledge is merged.	**Verification Protocol:** Public Metadata (category/colour) is merged with Private Metadata (Hidden Notes) and Logistical Metadata (Bin Mapping) into a relational database.
Internalization	Explicit knowledge drives action.	**Coordination & Resolution:** Users internalise Retrieval IDs to execute secure physical handovers. Data is archived to prevent Inventory Drift.

5. Core Features (MVP ONLY)

Feature	MVP Description	Priority
Categorical Icon Grid	A non-scrollable grid of SVG icons to standardise item classification (no free-text input).	Must Have
Fixed Colour Palette	An 8-button hex-code colour picker to standardise colour attributes.	Must Have
Zone Dropdown	Hard-coded spatial locations (e.g., "Main Library") to prevent disparate naming.	Must Have
Masked Chat Module	Ephemeral messaging window for Blind Verification. Conceals identities.	Must Have
Information Scrubbing	Regex-based middleware that scans and censors phone numbers and emails in chat.	Must Have
The “Hidden Note” Vault	An encrypted field for private identifiers used during Blind Verification.	Must Have
Retrieval ID Generator	Generates a 4-digit code (e.g., 883A) once a claim is approved for safe handover.	Must Have
Bin Mapping System	Matches digital records to physical barcodes/bins for staff logistics.	Must Have

6. App Pages

Page	Purpose	Main Content / Actions
1. Universal Knowledge Board	Public Discovery of found items.	Filtered list of items utilising modern, clean horizontal card layouts. Native browser alerts are utilised for simple interaction prompts.
2. Item Submission Interface	Structured intake of new items.	Icon Grid Picker, Fixed Colour Palette, Zone Dropdown, Image Upload (with PII disclaimer), Hidden Note input.
3. Masked Chat Interface	Secure communication for Blind Verification	Ephemeral messaging UI, Information Scrubbing active, "Approve Handover" or "Reject Claim" actions.
4. Safe Drop Coordination	Finalise the physical transfer.	Displays Retrieval ID and lists verified physical drop-off locations.

5. **Staff/Admin Dashboard**	Operational control and logistics.	Bin Mapping, Audit Logs, Lifecycle Management flags (30-day alerts), Hidden Note decryption.

7. User Flow

Phase 1: Item Discovery and Initiation

1. Finder logs an item using the fixed taxonomy and adds a Private Tier "Hidden Note."
2. Claimant browses the Universal Knowledge Board, identifies a match, and clicks *Initiate Retrieval*.
3. Claimant selects a privacy state (Public or Anonymous).

Phase 2: The Verification Protocol (Blind Verification)

1. A masked chat thread opens.
2. Claimant must accurately describe an unlisted detail matching the "Hidden Note."
3. The holding party clicks *Approve for Handover*.

Phase 3: Coordination and Item Handover

1. **Scenario A (Mutual Public):** Direct meetup in a safe public location.
2. **Scenario B (Anonymous Proxy):** The holding party drops the item at a Safe Drop Point. A Retrieval ID is generated for the Claimant to present upon pickup.

Phase 4: Resolution and Archiving

1. Both users confirm the transfer by clicking *Handover Complete*.
2. The system updates the status to "Resolved," removing it from the board and archiving the chat logs.

8. Basic Database Structure

Table	Purpose	Key Fields
profiles	Store user identity and privacy state.	id, role, username, is_anonymous, created_at
items	Core knowledge objects.	post_id, category_id, color_hex, zone_id, hidden_note, status, bin_number, creator_id
categories	Icon-based taxonomy.	id, name, svg_url, is_active
zones	Spatial logic constraints.	id, name, description

chats	Masked communication threads.	id, post_id, claimant_id, status, retrieval_id
audit_logs	Accountability tracking.	id, post_id, staff_id, old_status, new_status, timestamp

9. MVP Scope (IMPORTANT)

Include in MVP	Exclude for Now
● Categorical Logic (Icon Grid, Fixed Colours) ● Masked Chat with Info Scrubbing ● Blind Verification Workflow ● Retrieval ID Generation ● Bin Mapping & Audit Trailing ● Public vs. Anonymous User States ● Modern Card-Based UI Layouts	● Computer Vision / AI Auto-tagging ● Proactive Geofencing Notifications ● NLP Verification Confidence Scores ● Complex Analytics Dashboards

10. Suggested Tech Stack

Layer	Technology
Frontend	Next.js, React, Tailwind CSS
Language	TypeScript
Backend / Database	Supabase PostgreSQL
Authentication	Supabase Auth
Storage	Supabase Storage (for item images)
Security / Parsing	Regex algorithms (for Information Scrubbing)

11. Future Features (Post-MVP)

- **Computer Vision (AI) Integration:** Auto-suggesting categories and colors based on uploaded images, and auto-blurring PII (faces/IDs).
- **Proactive Geofencing:** Sending push alerts to users who were in a specific spatial zone when an item was reported found.
- **NLP Verification Assistance:** Scanning Masked Chat text against the Hidden Note to provide Staff with a "Confidence Score" prior to handover approval.

12. 6-Week Sprint Delivery Alignment

Optimised for a 5-member student development team.

Sprint	Focus	Expected Working Outcome
Sprint 1 - Weeks 1-2	Schema setup, UI Foundations, Fixed Choice Taxonomy.	Database is live. Users can submit items using the card-based UI, Icon Grid, Fixed Colours, and Zones.
Sprint 2 - Weeks 3-4	Masked Chat, Blind Verification, Info Scrubbing.	Claimants can initiate retrieval. Masked chat censors PII, allowing for secure verification.
Sprint 3 - Weeks 5-6	Staff Dashboard, Logistics, Audit Logs, QA.	Staff can map physical bins. Retrieval IDs generate properly. System flags 30-day old items.

13. Admin and Staff Access

Role	Access Scope	Key Permissions
Staff	Operational logistics and verification.	Map physical bins, view Hidden Notes, monitor audit logs, approve/reject handover statuses.
Admin	Platform governance and taxonomy control.	All Staff permissions, plus modifying the Categorical Icon Grid, blocking malicious users, and enforcing lifecycle management (purging 30-day records).